

Exercise -3 (PYQ's)

Expressing Concentrations of Solutions

1. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): 3.1500g of hydrated oxalic acid dissolved in water to make 250.0 mL solution will result in 0.1 M oxalic acid solution.

Reason (R): Molar mass of hydrated oxalic acid is 126 g mol^{-1} . [10 April, 2023 (Shift-II)]

In the light of the above statements, choose the correct answer from the options given below:

- (a) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (b) (A) is false but (R) is true.
 (c) (A) is true but (R) is false.
 (d) Both (A) and (R) are true and (R) is the correct explanation of (A).
2. What is the mass ratio of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$, molar mass = 62 g/mol) required for making 500 g of 0.25 molal aqueous solution and 250 mL of 0.25 molar aqueous solution? [25 Jan, 2023 (Shift-II)]
 (a) 1 : 1 (b) 3 : 1
 (c) 2 : 1 (d) 1 : 2
3. A solution is prepared by adding 2g of "X" of 1 mole of water. Mass percent of "X" in the solution is : [11 April, 2023 (Shift-II)]
 (a) 20% (b) 5%
 (c) 2% (d) 10%
4. The density of 3 M solution of NaCl is 1.0 g mL^{-1} . Molality of the solution is $\times 10^{-2} \text{ m}$. (Nearest Integer). Given: Molar mass of Na and Cl is 23 and 35.5 g mol^{-1} respectively. [1 Feb, 2023 (Shift-I)]
5. 5g of NaOH was dissolved in deionized water to prepare a 450mL stock solution. What volume (in mL) of this solution would be required to prepare 500mL of 0.1 M solution? Given: Molar Mass of Na, O and H is 23, 16 and 1 g mol^{-1} respectively [24 Jan, 2023 (Shift-I)]
6. The number of units, which are used to express concentration of solutions from the following is _____. Mass percent, Mole, Mole fraction, Molarity, ppm, Molality. [24 Jan, 2023 (Shift-II)]
7. The molality of a 10%(V/V) solution of di-bromine solution in CCl_4 (carbon tetrachloride) is 'x'. $x = \times 10^{-2} \text{ M}$. (Nearest Integer) [Given: molar mass of $\text{Br}_2 = 160 \text{ g mol}^{-1}$ atomic mass of C = 12 g mol^{-1}

atomic mass of Cl = 35.5 g mol^{-1}

density of dibromine 3.2 g cm^{-3}

density of $\text{CCl}_4 = 1.6 \text{ g cm}^{-3}$

[1 Feb, 2023 (Shift-II)]

8. A solution of sugar is obtained by mixing 200 g of its 25% solution and 500 g of its 40% solution (both by mass). The mass percentage of the resulting sugar solution is. (Nearest Integer) [11 April, 2023 (Shift-I)]
9. The volume of HCl, containing 73 g L^{-1} , required to completely neutralise NaOH obtained by reacting 0.69 g of metallic sodium with water, is _____ mL. (Nearest Integer) (Given : molar Masses of Na, Cl, O, H are 23, 35.5, 16 and 1 g mol^{-1} respectively) [29 Jan, 2023 (Shift-II)]
10. Some amount of dichloromethane (CH_2Cl_2) is added to 671.141 mL of chloroform (CHCl_3) to prepare $2.6 \times 10^{-3} \text{ M}$ solution of CH_2Cl_2 (DCM). The concentration of DCM is _____ ppm (by mass). Given: Atomic mass: C = 12; H = 1; Cl = 35.5 density of $\text{CHCl}_3 = 1.49 \text{ g cm}^{-3}$ [30 Jan, 2023 (Shift-I)]
11. A commercially sold conc. HCl is 35% HCl by mass. If the density of this commercial acid is 1.46 g / mL , the molarity of this solution is: (Atomic mass: Cl = 35.5 amu, H = 1 amu) [26 June, 2022 (Shift-I)]
 (a) 10.2 M
 (b) 12.5 M
 (c) 14.0 M
 (d) 18.2 M

12. **Assertion (A):** At 10°C , the density of a 5 M solution of KCl [atomic masses of K and Cl are 39 and 35.5 g mol^{-1} respectively is 'x' g mL^{-1} . The solution is cooled to -21°C . The molality of the solution will remain unchanged.

Reason (R): The molality of a solution does not change with temperature as mass remains unaffected with temperature.

In the light of the above statements, choose the correct answer from the options given below

[27 June, 2022 (Shift-I)]

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 (c) (A) is true but (R) is false.
 (d) (A) is false but (R) is true.
13. When 800 mL of 0.5 M nitric acid is heated in a beaker, its volume is reduced to half and 11.5 g of nitric acid is evaporated. The molarity of the remaining nitric acid solution is $x \times 10^{-2} \text{ M}$. (Nearest Integer) (Molar mass of nitric acid is 63 g mol^{-1})

[26 July, 2022 (Shift-I)]

14. The molarity of the solution prepared by dissolving 6.3 g of oxalic acid ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$) in 250 mL of water in mol L^{-1} is $x \times 10^{-2}$. The value of x is _____. (Nearest Integer)
[Atomic mass : H : 1.0, C : 12.0, O : 16.0]
[31 Aug, 2021 (Shift-I)]
15. If 80 g of copper sulphate $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is dissolved in deionised water to make 5 L of solution. The concentration of the copper sulphate solution is $x \times 10^{-3} \text{ mol L}^{-1}$. The value of x is _____.
[Atomic masses Cu : 63.54 u, S : 32 u, O : 16 u, H : 1 u]
[1 Sept, 2021 (Shift-II)]
16. A 6.50 molal solution of KOH (aq) has a density of 1.89 g cm^{-3} . The molarity of the solution is _____ mol dm^{-3} . (Round off to the nearest Integer).
[Atomic masses : K = 39.0u, O = 16.0u, H = 1.0u]
[16 March, 2021 (Shift-I)]
17. A 1 molal $\text{K}_4\text{Fe}(\text{CN})_6$ solution has a degree of dissociation of 0.4. Its boiling point is equal to that of another solution which contains 18.1 weight percent of a non electrolytic solute A. The molar mass of A is _____ u. (Round off to the Nearest Integer). [Density of water = 1.0 g cm^{-3}]
[17 March, 2021 (Shift-II)]
18. An aqueous KCl solution of density 1.2 g mL^{-1} has a molality of 3.30 mol kg^{-1} . The molarity of the solution in mol L^{-1} is _____. (Nearest Integer)
[Molar mass of KCl = 74.5] **[26 Aug, 2021 (Shift-I)]**
19. 4.5 g of compound A (MW = 90) was used to make 250 mL of its aqueous solution. The molarity of the solution in M is $x \times 10^{-1}$. The value of x is _____. (Rounded off to the nearest Integer) **[24 Feb, 2021 (Shift-I)]**
20. The mole fraction of a solute in a 100 molal aqueous solution is _____ $\times 10^{-2}$ (Round off to the Nearest Integer).
[Given: Atomic masses: H: 1.0 u, O: 16.0 u]
[17 March, 2021 (Shift-I)]
21. The density of NaOH solution is 1.2 gm cm^{-3} . The molality of this solution is _____ m.
(Round off to the Nearest Integer)
[Use: Atomic masses : Na 23.0u O: 16.0 u H: 1.0 u Density of H_2O : 1.0 g cm^{-3}]
[27 July, 2021 (Shift-I)]
22. Sodium oxide reacts with water to produce sodium hydroxide. 20.0 g of sodium oxide is dissolved in 500 mL of water. Neglecting the change in volume, the concentration of the resulting NaOH solution is _____ $\times 10^{-1} \text{ M}$. (Nearest Integer)
[Atomic mass : Na = 23.0, O = 16.0, H = 1.0]
[31 Aug, 2021 (Shift-II)]
23. If the concentration of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) in blood is 0.72 g L^{-1} , the molarity of glucose in blood is _____ $\times 10^{-3} \text{ M}$. (Nearest Integer)
(Given : Atomic mass of C = 12, H = 1, O = 16 u)
[22 July, 2021 (Shift-II)]
24. 100 mL of Na_3PO_4 solution contains 3.45 g of sodium. The molarity of the solution is _____ $\times 10^{-2} \text{ mol L}^{-1}$, (Nearest Integer)
[Atomic Masses – Na : 23.0 u, O : 16.0 u, P : 31.0 u]
[26 Aug, 2021 (Shift-II)]
25. A solution of two components containing n_1 moles of the 1st component and n_2 moles of the 2nd component is prepared. M_1 and M_2 are the molecular weights of component 1 and 2 respectively. If d is the density of the solution in gmL^{-1} , C_2 is the molarity and x_2 is the mole fraction of the 2nd component, then C_2 can be expressed as: **[6 Sept, 2020 (Shift-I)]**
- (a) $C_2 = \frac{1000x_2}{M_1 + x_2(M_2 - M_1)}$
- (b) $C_2 = \frac{dx_2}{M_2 + x_2(M_2 - M_1)}$
- (c) $C_2 = \frac{1000dx_2}{M_1 + x_2(M_2 - M_1)}$
- (d) $C_2 = \frac{dx_1}{M_2 + x_2(M_2 - M_1)}$
26. The molarity of HNO_3 in a sample which has density 1.4 g/mL and mass percentage of 63% is (Molecular Weight of $\text{HNO}_3 = 63$) **[9 Jan, 2020 (Shift-I)]**
27. A 100 mL solution was made by adding 1.43 g of $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$. The normality of the solution is 0.1 N. The value of x is _____.
(The atomic mass of Na is 23 g/mol)
[4 Sept, 2020 (Shift-II)]
28. The hardness of a water sample containing 10^{-3} M MgSO_4 expressed as CaCO_3 equivalents (in ppm) is _____.
(molar mass of MgSO_4 is 120.38 g/mol)
[9 Jan, 2020 (Shift-I)]
29. 10.30 mg of O_2 is dissolved into a litre of sea water of density 1.03 g/mL . The concentration of O_2 in ppm is _____.
[9 Jan, 2020 (Shift-II)]
30. The mole fraction of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) in an aqueous binary solution is 0.1. The mass percentage of water in it, to the nearest Integer, is _____.
[3 Sept, 2020 (Shift-I)]
31. 6.023×10^{22} molecules are present in 10 g of a substance 'x'. The molarity of a solution containing 5 g of substance 'x' in 2 L solution is _____ $\times 10^{-3}$.
[3 Sept, 2020 (Shift-II)]

32. 8 g of NaOH is dissolved in 18 g of H_2O . Mole fraction of NaOH in solution and molality (in mol kg^{-1}) of the solution respectively, are: [12 Jan, 2019 (Shift-II)]
 (a) 0.2, 22.20 (b) 0.2, 11.11
 (c) 0.167, 11.11 (d) 0.167, 22.20
33. The mole fraction of a solvent in aqueous solution of a solute is 0.8. The molality (in mol kg^{-1}) of the aqueous solution is: [12 Jan, 2019 (Shift-II)]
 (a) 13.88×10^{-1} (b) 13.88×10^{-2}
 (c) 13.88 (d) 13.88×10^{-3}
34. The amount of sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) required to prepare 2 L of its 0.1 M aqueous solution is: [10 Jan, 2019 (Shift-II)]
 (a) 136.8 g (b) 17.1 g
 (c) 68.4 g (d) 34.2 g
35. A solution of sodium sulphate contains 92 g of Na^+ ions per kilogram of water. The molality of Na^+ ions in that solution in mol kg^{-1} is: [9 Jan, 2019 (Shift-I)]
 (a) 12 (b) 4
 (c) 8 (d) 16
36. Liquids A and B form an ideal solution in the entire composition range. At 350 K, the vapor pressures of pure A and pure B are 7×10^3 Pa and 12×10^3 Pa, respectively. The composition of the vapour in equilibrium with a solution containing 40 mole percent of A at this temperature is [10 Jan, 2019 (Shift-I)]
 (a) $X_A = 0.76$; $X_B = 0.24$ (b) $X_A = 0.37$; $X_B = 0.63$
 (c) $X_A = 0.28$; $X_B = 0.72$ (d) $X_A = 0.4$; $X_B = 0.6$
37. What would be the molality of 20% (mass/mass) aqueous solution of KI? (molar mass of KI = 166 g mol^{-1}) [9 April, 2019 (Shift-II)]
 (a) 1.08 (b) 1.48
 (c) 1.51 (d) 1.35
38. The mole fraction of urea in an aqueous urea solution containing 900 g of water is 0.05. If the density of the solution is 1.2 g cm^{-3} , the molarity of urea solution is _____. (Given data: Molar masses of urea and water are 60 g mol^{-1} and 18 g mol^{-1} , respectively) [JEE Adv (2019)]

Solubility and Vapour Pressure of Liquid Solutions

39. Solid Lead nitrate is dissolved in 1 litre of water. The solution was found to boil at 100.15°C . When 0.2 mol of NaCl is added to the resulting solution, it was observed that the solution froze at -0.8°C . The solubility product of PbCl_2 formed is _____ $\times 10^{-6}$ at 298 K.

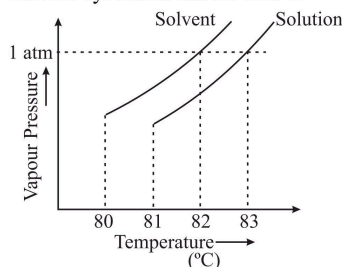
(Nearest Integer)

Given: $K_b = 0.5 \text{ K kg mol}^{-1}$ and $K_f = 1.8 \text{ K kg mol}^{-1}$. Assume molality to be equal to molarity in all cases.

[29 Jan, 2023 (Shift-I)]

40. Mass of Urea (NH_2CONH_2) required to be dissolved in 1000 g of water to reduce the vapour pressure of water by 25% is _____ g. (Nearest Integer)
 Given: Molar mass of N, C, O and H are 14, 12, 16 and 1 g mol^{-1} respectively. [6 April, 2023 (Shift-I)]

41. The vapour pressure vs. temperature curve for a solution solvent system is shown below.



The boiling point of the solvent is $^\circ\text{C}$

[8 April, 2023 (Shift-I)]

42. 80 mole percent of MgCl_2 is dissociated in aqueous solution. The vapour pressure of 1.0 molal aqueous solution of MgCl_2 at 38°C is _____ mm Hg. (Nearest Integer)
 Given: Vapour pressure of water at 38°C is 50 mmHg [12 April, 2023 (Shift-I)]
43. The Total pressure observed by mixing two liquid A and B is 350 mm Hg when their mole fractions are 0.7 and 0.3 respectively.
 The Total pressure becomes 410 mm Hg if the mole fractions are changed to 0.2 and 0.8 respectively for A and B. The vapour pressure of pure A is mmHg. (Nearest Integer)
 Consider the liquids and solutions behave ideally.

[24 Jan, 2023 (Shift-II)]

44. The total pressure of a mixture of non-reacting gases X (0.6 g) and Y (0.45 g) in a vessel is 740 mm of Hg. The partial pressure of the gas X is _____ mm of Hg. (Nearest Integer)
 (Given: molar mass X = 20 and Y = 45 g mol^{-1}) [31 Jan, 2023 (Shift-I)]
45. A company dissolves 'X' amount of CO_2 at 298 K in 1 litre of water to prepare soda water. $X = ______ \times 10^{-3} \text{ g}$. (nearest Integer)

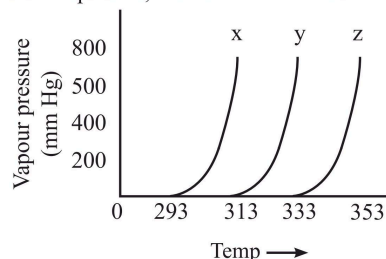
(Given: partial pressure of CO_2 at 298 K = 0.835 bar. Henry's law constant for CO_2 at 298 K = 1.67 kbar. Atomic mass of H, C and O is 1, 12 and 16 g mol^{-1} , respectively)

[24 June, 2022 (Shift-II)]

46. A gaseous mixture of two substances A and B, under a total pressure of 0.8 atm is in equilibrium with an ideal liquid solution. The mole fraction of substance A is 0.5 in the vapour phase and 0.2 in the liquid phase. The vapour pressure of pure liquid A is _____ atm. (Nearest Integer) [28 July, 2022 (Shift-II)]
47. The vapour pressure of two volatile liquids A and B at 25°C are 50 Torr and 100 Torr, respectively. If the liquid mixture contains 0.3 mole fraction of A, then the mole fraction of liquid B in the vapour phase is $\frac{x}{17}$. The value of x is _____. [28 June, 2022 (Shift-II)]
48. If O_2 gas is bubbled through water at 303 K, the number of millimoles of O_2 gas that dissolve in 1 litre of water is _____. (Nearest Integer)
(Given: Henry's Law constant for O_2 at 303 K is 46.82 k bar and partial pressure of $O_2 = 0.920$ bar)
(Assume solubility of O_2 in water is too small, nearly negligible) [29 July, 2022 (Shift-II)]
49. The oxygen dissolved in water exerts a partial pressure of 20 kPa in the vapour above water. The molar solubility of oxygen in water is $\times 10^{-5}$ mol dm^{-3} . (Round off to the Nearest Integer).
[Given: Henry's law constant = $K_H = 8.0 \times 10^4$ kPa for O_2
Density of water with dissolved oxygen = 1.0 kg dm^{-3}] [17 March, 2021 (Shift-I)]
50. CO_2 gas is bubbled through water during a soft drink manufacturing process at 298 K. If CO_2 exerts a partial pressure of 0.835 bar, then x mol of CO_2 would dissolve in 0.9 L of water. The value of x is _____. (Nearest Integer)
(Henry's law constant for CO_2 at 298 K is 1.67×10^3 bar) [25 July, 2021 (Shift-I)]
51. At 20°C, the vapour pressure of benzene is 70 torr and that of methyl benzene is 20 torr. The mole fraction of benzene in the vapour phase at 20°C above an equimolar mixture of benzene and methyl benzene is $\times 10^{-2}$. (Nearest Integer) [20 July, 2021 (Shift-I)]
52. The vapour pressures of A and B at 25°C are 90 mm Hg and 15 mm Hg respectively. If A and B are mixed such that the mole fraction of A in the mixture is 0.6, then the mole fraction of B in the vapour phase is $x \times 10^{-1}$. The value of x is _____. (Nearest Integer) [20 July, 2021 (Shift-II)]
53. At 35°C, the vapour pressure of CS_2 is 512 mm of Hg and that of acetone is 344 mm of Hg. A solution of CS_2 in acetone has a total vapour pressure of 600 mm of Hg. The false statement among the following is
(a) CS_2 and acetone are less attracted to each other than to themselves

- (b) Heat must be absorbed in order to produce the solution at 35°C
(c) Raoult's law is not obeyed by this system
(d) a mixture of 100 mL CS_2 and 100 mL acetone has a volume of < 200 mL [7 Jan, 2020 (Shift-I)]

54. A graph of vapour pressure and temperature for three different liquids X, Y and Z is shown below.



The following inferences are made

- (A) X has higher intermolecular interactions as compared to Y.
(B) X has lower intermolecular interactions as compared to Y.
(C) Z has lower intermolecular interactions as compared to Y.

The correct inferences is/are [8 Jan, 2020 (Shift-I)]

- (a) (B)
(b) (C)
(c) (A) and (C)
(d) (A)

55. Henry's constant (in kbar) for four gases α , β , γ and δ in water at 298 K is given below:

	α	β	γ	δ
K_H	50	2	2×10^{-5}	0.5

(density of water = 10^3 kg m^{-3} at 298 K)

This table implies that:

- (a) α has the highest solubility in water at a given pressure
(b) solubility of γ at 308 K is lower than at 298 K.
(c) The pressure of a 55.5 molal solution of δ is 250 bar.
(d) The pressure of a 55.5 molal solution of γ is 1 bar.

[3 Sept, 2020 (Shift-I)]

56. An open beaker of water in equilibrium with its water vapour is in a sealed container. When a few grams of glucose are added to the beaker of water, the rate at which water molecules [2 Sept, 2020 (Shift-I)]

- (a) Leaves the solution increases
(b) Leaves the vapour increases
(c) Leaves the vapour decreases
(d) Leaves the solution decreases

57. At 300 K, the vapour pressure of a solution containing 1 mole of n-hexane and 3 moles of n-heptane is 550 mm of Hg. At the same temperature, if one more mole of n-heptane is added to this solution, the vapour pressure of the solution increases by 10 mm of Hg. What is the vapour pressure in mm Hg of n-heptane in its pure state?

[4 Sept, 2020 (Shift-I)]

58. Liquids A and B form an ideal solution for all compositions of A and B at 25°C. Two such solutions with 0.25 and 0.50 mole fractions of A have the total vapor pressures of 0.3 and 0.4 bar, respectively. What is the vapor pressure of pure liquid B in bar?

[JEE Adv 2020]

59. Liquid 'M' and liquid 'N' form an ideal solution. The vapour pressures of pure liquids 'M' and 'N' are 450 and 700 mm Hg, respectively, at the same temperature. Then, correct statement is

(X_M = Mole fraction of 'M' in solution;

X_N = Mole fraction of 'N' in solution;

y_M = Mole fraction of 'M' in vapour phase;

y_N = Mole fraction of 'N' in vapour phase)

[9 April, 2019 (Shift-I)]

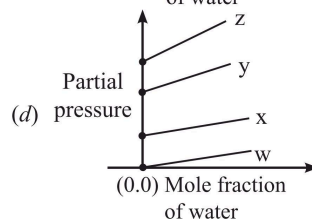
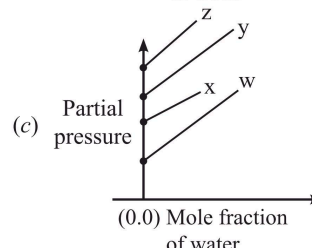
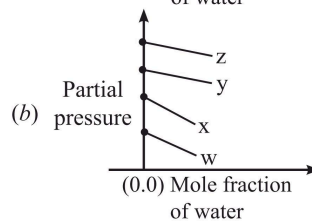
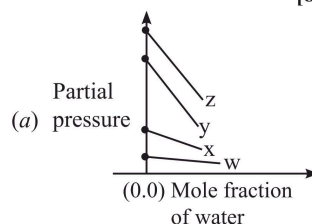
- (a) $\frac{X_M}{X_N} = \frac{y_M}{y_N}$ (b) $\frac{X_M}{X_N} > \frac{y_M}{y_N}$
- (c) $\frac{X_M}{X_N} < \frac{y_M}{y_N}$ (d) $(X_M - y_M) < (X_N - y_N)$
60. Which one of the following statements regarding Henry's law is not correct?
- [9 Jan, 2019 (Shift-I)]
- (a) Different gases have different K_H (Henry's law constant) values at the same temperature.
- (b) The value of K_H increases with increase of temperature and K_H is a function of the nature of the gas.
- (c) The partial pressure of the gas in vapour phase is proportional to the mole fraction of the gas in the solution.
- (d) Higher the value of K_H at a given pressure, higher is the solubility of the gas in the liquid.
61. The vapour pressures of pure liquids A and B are 400 and 600 mm Hg, respectively at 298 K. On mixing the two liquids, the sum of their initial volumes is equal to the volume of the final mixture. The mole fraction of liquid B is 0.5 in the mixture. The vapour pressure of the final solution, the mole fractions of components A and B in vapour phase respectively, are
- [8 April, 2019 (Shift-I)]
- (a) 500 mm Hg, 0.4, 0.6 (b) 500 mm Hg, 0.5, 0.5
- (c) 450 mm Hg, 0.4, 0.6 (d) 450 mm Hg, 0.5, 0.5
62. If solubility product of $Zr_3(PO_4)_4$ is denoted by K_{sp} and its molar solubility is denoted by S, then which of the following relation between S and K_{sp} is correct?

(a) $S = \left(\frac{K_{sp}}{929}\right)^{1/9}$ (b) $S = \left(\frac{K_{sp}}{216}\right)^{1/7}$

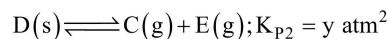
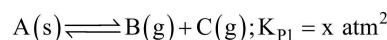
(c) $S = \left(\frac{K_{sp}}{144}\right)^{1/6}$ (d) $S = \left(\frac{K_{sp}}{6912}\right)^{1/7}$

63. For the solution of the gases w, x, y and z in water at 298 K, the Henry's law constants (K_H) are 0.5, 2, 35 and 40 k bar respectively. The correct plot for the given data is

[8 April, 2019 (Shift-II)]



64. Two solids dissociate as follows:



The total pressure when both the solids dissociate simultaneously is

[12 Jan, 2019 (Shift-I)]

- (a) $\sqrt{x + y} \text{ atm}$
- (b) $2(\sqrt{x + y}) \text{ atm}$
- (c) $(x + y) \text{ atm}$
- (d) $x^2 + y^2 \text{ atm}$

65. At room temperature, a dilute solution of urea is prepared by dissolving 0.60 g of urea in 360 g of water. If the vapour pressure of pure water at this temperature is 35 mm Hg, lowering of vapour pressure will be: (molar mass of urea = 60 g mol⁻¹) [10 April, 2019 (Shift-I)]

(a) 0.027 mmHg (b) 0.031 mmHg
(c) 0.028 mmHg (d) 0.017 mmHg

66. Liquids **A** and **B** form an ideal solution for all compositions of **A** and **B** at 25°C. Two such solutions with 0.25 and 0.50 mole fractions of **A** have the total vapor pressures of 0.3 and 0.4 bar, respectively. What is the vapor pressure of pure liquid **B** in bar? [JEE Adv (2020)]

Ideal and Non-Ideal Solutions

67. Two open beakers one containing a solvent and the other containing a mixture of that solvent with a non-volatile solute are together sealed in a container. Over time.
- (a) The volume of the solution and the solvent does not change.
(b) The volume of the solution increases and the volume of the solvent decreases.
(c) The volume of the solution does not change and the volume of the solvent decreases.
(d) The volume of the solution decreases and the volume of the solvent increases. [7 Jan, 2020 (Shift-II)]

Colligative Properties and Abnormal Molar Mass

68. Evaluate the following statements for their correctness.
- (A) The elevation in boiling point temperature of water will be same for 0.1 M NaCl and 0.1 M urea.
(B) Azeotropic mixtures boil without change in their composition
(C) Osmosis always takes place from hypertonic to hypotonic solution
(D) The density of 32% H₂SO₄ solution having molarity 4.09 M is approximately 1.26 g mL⁻¹
(E) A negatively charged sol is obtained when KI solution is added to silver nitrate solution.

Choose the correct answer from the options given below :

[31 Jan, 2023 (Shift-II)]

- (a) (B), (D), and (E) only (b) (A), (B), and (D) only
(c) (A) and (C) only (d) (B) and (D) only
69. What weight of glucose must be dissolved in 100 g of water to lower the vapour pressure by 0.20 mm Hg? (Assume dilute solution is being formed)
Given: Vapour pressure of pure water is 54.2 mm Hg at room temperature. Molar mass of glucose is 180 g mol⁻¹.

[11 April, 2023 (Shift-II)]

- (a) 4.69 g
(b) 3.59 g
(c) 2.59 g
(d) 3.69 g

70. In the depression of freezing point experiment

[24 Jan, 2023 (Shift-I)]

- (A) Vapour pressure of the solution is less than that of pure solvent
(B) Vapour pressure of the solution is more than that of pure solvent
(C) Only solute molecules solidify at the freezing point
(D) Only solvent molecules solidify at the freezing point
(a) (A) and (D) only
(b) (B) and (C) only
(c) (A) and (C) only
(d) (A) only

71. Match List-I with List-II. [29 Jan, 2023 (Shift-II)]

List-I

- (A) Van't Hoff Factor, i
(B) k_f
(C) Solutions with same
(D) Azeotropes

List-II

- (I) Cryoscopic constant
(II) Isotonic solutions
(III) Normal molar mass
Abnormal molar mass
(IV) Solutions with same composition of vapour above it

Choose the correct answer from the options given below:

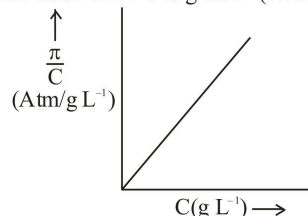
- (a) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)
(b) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)
(c) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
(d) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

72. Solution of 12 g of non-electrolyte (A) prepared by dissolving it in 1000 mL of water exerts the same osmotic pressure as that of 0.05 M glucose solution at the same temperature. The empirical formula of A is CH₂O. The molecular mass of A is _____ g. (Nearest Integer)

[13 April, 2023 (Shift-I)]

73. The osmotic pressure of solutions of PVC in cyclohexanone at 300 K are plotted on the graph.

The molar mass of PVC is g mol⁻¹ (Nearest Integer)



(Given : $R = 0.083 \text{ L atm K}^{-1} \text{ mol}^{-1}$)

[25 Jan, 2023 (Shift-I)]

74. The number of pairs of the solution having the same value of the osmotic pressure from the following is _____.
(Assume 100% ionization) [25 Jan, 2023 (Shift-II)]
(A) 0.500 M C_2H_5OH and 0.25 M KBr (aq)
(B) 0.100 M $K_4[Fe(CN)_6]$ (aq) and 0.100 M $FeSO_4(NH_4)_2SO_4$ (aq)
(C) 0.05 M $K_4[Fe(CN)_6]$ (aq) and 0.25 M $NaCl$ (aq)
(D) 0.15 M $NaCl$ (aq) and 0.1 M $BaCl_2$ (aq)
(E) 0.02 M KCl $MgCl_2 \cdot 6H_2O$ (aq) and 0.05 M KCl (aq)
75. A solution containing 2g of a non-volatile solute in 20g of water boils at 373.52 K. The molecular mass of the solute is $gmol^{-1}$. (Nearest Integer)
Given, water boils at 373 K, K_b for water = 0.52 K kg mol^{-1}
[30 Jan, 2023 (Shift-I)]
76. Consider the following pairs of solution which will be isotonic at the same temperature. The number of pairs of solutions is/are. [6 April, 2023 (Shift-II)]
(a) 1 M aq. $NaCl$ and 2 M aq. Urea
(b) 1 M aq. $CaCl_2$ and 1.5 M aq. KCl
(c) 1.5 M aq. $AlCl_3$ and 2 M aq. Na_2SO_4
(d) 2.5 M aq. KCl and 1 M aq. $Al_2(SO_4)_3$
77. Lead storage battery contains 38% by weight solution of H_2SO_4 . The van't Hoff factor is 2.67 at this concentration. The temperature in Kelvin at which the solution in the battery will freeze is (Nearest Integer).
Given $K_f = 1.8 K kg mol^{-1}$ [30 Jan, 2023 (Shift-II)]
78. 20% of acetic acid is dissociated when its 5g is added to 500mL of water. The depression in freezing point of such water is $\times 10^{-3}^\circ C$. Atomic mass of C, H and O are 12, 1 and 16 a.m.u. respectively.
[Given: Molal depression constant and density of water are 1.86 K kg mol^{-1} and 1 g cm^{-3} respectively].
[1 Feb, 2023 (Shift-II)]
79. At 27°C, a solution containing 2.5 g of solute in 250.0 mL of solution exerts an osmotic pressure of 400 Pa. The molar mass of the solute is ____ g/mol (Nearest Integer)
(Given : $R = 0.083 L bar K^{-1} mol^{-1}$)
[31 Jan, 2023 (Shift-I)]
80. If the boiling points of two solvents X and Y (having same molecular weights) are in the ratio 2 : 1 and their enthalpy of vaporizations are in the ratio 1 : 2, then the boiling point elevation constant of X is m times the boiling point elevation constant of Y. The value of m is ____ (Nearest Integer) [8 April, 2023 (Shift-II)]
81. An aqueous solution of volume 300 cm^3 contains 0.63 g of protein. The osmotic pressure of the solution at 300 K is 1.29 mbar. The molar mass of the protein is ____ g mol^{-1}
Given : $R = 0.083 L bar K^{-1} mol^{-1}$ [8 April, 2023 (Shift-II)]
82. The vapour pressure of 30 % (w/v) aqueous solution of glucose is _____ mm Hg at 25°C.

[Given : The density of 30%(w/v), aqueous solution of glucose is 1.2 g cm^{-3} and vapour pressure of pure water is 24 mmHg.]

(Molar mass of glucose is 180 g mol^{-1} .)

[15 April, 2023 (Shift-I)]

83. If the degree of dissociation of aqueous solution of weak monobasic acid is determined to be 0.3, then the observed freezing point will be _____ % higher than the expected/theoretical freezing point. (Nearest Integer)

[10 April, 2023 (Shift-I)]

84. 0.004 M K_2SO_4 solution is isotonic with 0.01 M glucose solution. Percentage dissociation of K_2SO_4 is ____ (Nearest Integer) [11 April, 2023 (Shift-I)]

85. Sea water contains 29.25% $NaCl$ and 19% $MgCl_2$ by weight of solution. The normal boiling point of the sea water is $^\circ C$ (Nearest Integer)

Assume 100% ionization for both $NaCl$ and $MgCl_2$

Given: $K_b (H_2O) = 0.52 K kg mol^{-1}$

Molar mass of $NaCl$ and $MgCl_2$ is 58.5 and 95 g mol^{-1} respectively. [13 April, 2023 (Shift-II)]

86. 50 mL of 0.2 molal urea solution (density = 1.012 g mL^{-1} at 300 K) is mixed with 250 mL of a solution containing 0.06 g of urea. Both the solutions were prepared in the same solvent. The osmotic pressure (in Torr) of the resulting solution at 300 K is ____.

[Use : Molar mass of urea = 60 g mol^{-1} ; gas constant, $R = 62 L Torr K^{-1} mol^{-1}$; Assume, $\Delta_{mix} H = 0$, $\Delta_{mix} V = 0$]

[4 June, 2023 (Shift-II)]

87. Two solutions A and B are prepared by dissolving 1 g of non-volatile solutes X and Y, respectively in 1 kg of water. The ratio of depression in freezing points for A and B is found to be 1:4. The ratio of molar masses of X and Y is

[25 July, 2022 (Shift-II)]

- (a) 1:4 (b) 1:0.25
(c) 1:0.20 (d) 1:5

88. Solute A associates in water. When 0.7 g of solute A is dissolved in 42.0 g of water, it depresses the freezing point by 0.2°C. The percentage association of solute A in water, is
[Given: Molar mass of A = 93 g mol^{-1} . Molal depression constant of water is 1.86 K kg mol^{-1}]

[25 June, 2022 (Shift-II)]

- (a) 50%
(b) 60%
(c) 70%
(d) 80%

89. The depression in freezing point observed for a formic acid solution of concentration 0.5 mol L^{-1} is 0.0405°C . Density of formic acid is 1.05 g mL^{-1} . The Van't Hoff factor of the formic acid solution is nearly: (Given for water $K_f = 1.86 \text{ K kg mol}^{-1}$) [25 July, 2022 (Shift-I)]
 (a) 0.8 (b) 1.1
 (c) 1.9 (d) 2.4
90. Boiling point of a 2% aqueous solution of a non-volatile solute A is equal to the boiling point of 8% aqueous of a non-volatile solute B. The relation between molecular weights of A and B is [27 July, 2022 (Shift-I)]
 (a) $M_A = 4M_B$ (b) $M_B = 4M_A$
 (c) $M_A = 8M_B$ (d) $M_B = 8M_A$
91. A 0.5 percent solution of potassium chloride was found to freeze at -0.24°C . The percentage dissociation of potassium chloride is _____. (Nearest Integer)
 (Molal depression constant for water is $1.80 \text{ K kg mol}^{-1}$ and molar mass of KCl is 74.6 g mol^{-1}) [26 June, 2022 (Shift-I)]
92. The osmotic pressure of blood is 7.47 bar at 300 K. To inject glucose to a patient intravenously, it has to be isotonic with blood. The concentration of glucose solution in g L^{-1} is _____. (Molar mass of glucose = 180 g mol^{-1} ($R = 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1}$)) (Nearest Integer)
 [24 June, 2022 (Shift-I)]
93. 2 g of a non-volatile non-electrolyte solute is dissolved in 200 g of two different solvents A and B whose ebullioscopic constants are in the ratio of 1 : 8. The elevation in boiling points of A and B are in the ratio of x/y ($x : y$). The value of y is _____. (Nearest Integer). [27 June, 2022 (Shift-I)]
94. 1.2 mL of acetic acid is dissolved in water to make 2.0 L of solution. The depression in freezing point observed for this strength of acid is 0.0198°C . The percentage of dissociation of the acid is _____. (Nearest Integer)
 [Given: Density of acetic acid is 1.02 g mL^{-1} Molar mass of acetic acid is 60 g mol^{-1} , $K_f(\text{H}_2\text{O}) = 1.85 \text{ K kg mol}^{-1}$] [29 June, 2022 (Shift-I)]
95. Elevation in boiling point for 1.5 molal solution of glucose in water is 4 K. The depression in freezing point for 4.5 molal solution of glucose in water is 4 K. The ratio of molal elevation constant to molal depression constant (K_b/K_f) is _____. [29 June, 2022 (Shift-II)]
96. 1.80 g of solute A was dissolved in 62.5 cm^3 of ethanol and freezing point of the solution was found to be 155.1 K . The molar mass of solute A is _____. g mol^{-1} .
 [Given: Freezing point of ethanol is 156.0 K . Density of ethanol is 0.80 g cm^{-3} . Freezing point depression constant of ethanol is $2.00 \text{ K kg mol}^{-1}$] [29 July, 2022 (Shift-II)]
97. The elevation in boiling point for 1 molal solution of non-volatile solute A is 3K. The depression in freezing point for 2 molal solution of A in the same solvent is 6 K. The ratio of K_b and K_f i.e. K_b/K_f is 1 : X. The value of X is _____. [Nearest Integer] [26 July, 2022 (Shift-II)]
98. When a certain amount of solid A is dissolved in 100 g of water at 25°C to make a dilute solution, the vapour pressure of the solution is reduced to one-half of that of pure water. The vapour pressure of pure water is 23.76 mmHg. The number of moles of solute A added is _____. (Nearest Integer) [27 July, 2022 (Shift-II)]
99. 150 g of acetic acid was contaminated with 10.2 g ascorbic acid ($\text{C}_6\text{H}_8\text{O}_6$) to lower down its freezing point by $(x \times 10^{-1})^\circ\text{C}$. The value of x is _____. (Nearest Integer)
 (Given $K_f = 3.9 \text{ K kg mol}^{-1}$; molar mass of ascorbic acid = 176 g mol^{-1}) [28 July, 2022 (Shift-I)]
100. The osmotic pressure exerted by a solution prepared by dissolving 2.0 g of protein of molar mass 60 kg mol^{-1} in 200 mL of water at 27°C is _____ Pa. (nearest Integer)
 (Use $R = 0.083 \text{ L bar mol}^{-1} \text{ K}^{-1}$) [26 June, 2022 (Shift-II)]
101. Which one of the following 0.06 M aqueous solutions has lowest freezing point? [22 July, 2021 (Shift-II)]
 (a) KI (b) $\text{Al}_2(\text{SO}_4)_3$
 (c) $\text{C}_6\text{H}_{12}\text{O}_6$ (d) K_2SO_4
102. Which one of the following 0.10 M aqueous solutions will exhibit the largest freezing point depression? [31 Aug, 2021 (Shift-I)]
 (a) Glycine (b) KHSO_4
 (c) Hydrazine (d) Glucose
103. A solute A dimerizes in water. The boiling point of a 2 molal solution of A is 100.52°C . The percentage association of A is _____. (Round off to the Nearest Integer).
 [Use: K_b for water = $0.52 \text{ K kg mol}^{-1}$ Boiling point of water = 100°C] [18 March, 2021 (Shift-II)]
104. In a solvent, 50% of an acid HA dimerizes and the rest dissociates. The van't Hoff factor of the acid is _____ $\times 10^{-2}$ (Round off to the Nearest Integer). [27 July, 2021 (Shift-II)]
105. Of the following four aqueous solutions, total number of those solutions whose freezing point is lower than that of 0.10 M $\text{C}_2\text{H}_5\text{OH}$ is _____. (Integer answer) [26 Aug, 2021 (Shift-I)]
 (a) 0.10 M $\text{Ba}_3(\text{PO}_4)_2$ (b) 0.10 M Na_2SO_4
 (c) 0.10 M KCl (d) 0.10 M Li_3PO_4
106. 83 g of ethylene glycol dissolved in 625 g of water. The freezing point of the solution is _____. (Nearest Integer)
 [Use : Molal freezing point depression constant of water = $1.86 \text{ K kg mol}^{-1}$, Freezing point of water = 273 K , Atomic masses: C: 12.0 u, O: 16.0 u, H: 1.0 u] [26 Aug, 2021 (Shift-II)]

107. When 12.2 g of benzoic acid is dissolved in 100 g of water, the freezing point of solution was found to be -0.93°C [$K_f(\text{H}_2\text{O}) = 1.86 \text{ K kg mol}^{-1}$]. The number (n) of benzoic acid molecules associated (assuming 100% association) is _____. [26 Feb, 2021 (Shift-II)]
108. 1 kg of 0.75 molal aqueous solution of sucrose can be cooled up to -4°C before freezing. The amount of ice (in g) that will be separated out is _____. (Nearest Integer) [Given: $K_f(\text{H}_2\text{O}) = 1.86 \text{ K kg mol}^{-1}$] [27 Aug, 2021 (Shift-I)]
109. When 9.45 g of ClCH_2COOH is added to 500 mL of water, its freezing point drops by 0.5°C . The dissociation constant of ClCH_2COOH is $x \times 10^{-3}$. The value of x is _____. (Rounded off to the nearest Integer) [$K_f(\text{H}_2\text{O}) = 1.86 \text{ K kg mol}^{-1}$] [24 Feb, 2021 (Shift-I)]
110. If a compound AB dissociates to the extent of 75% in an aqueous solution, the molality of the solution which shows a 2.5 K rise in the boiling point of the solution, is _____ molal. (Rounded-off to the nearest Integer) [$K_b = 0.52 \text{ K kg mol}^{-1}$] [25 Feb, 2021 (Shift-II)]
111. 1.46 g of a biopolymer dissolved in a 100 mL water at 300 K exerted an osmotic pressure of 2.42×10^{-3} bar. The molar mass of the biopolymer is _____ $\times 10^4 \text{ g mol}^{-1}$ (Round off to the Nearest Integer) [Use: $R = 0.083 \text{ L bar mol}^{-1} \text{ K}^{-1}$] [27 July, 2021 (Shift-I)]
112. 1 molal aqueous solution of an electrolyte A_2B_3 is 60% ionised. The boiling point of the solution at 1 atm is _____ K. (Rounded-off to the nearest Integer) [Given: K_b for $(\text{H}_2\text{O}) = 0.52 \text{ K kg mol}^{-1}$] [25 Feb, 2021 (Shift-I)]
113. AB_2 is 10% dissociated in water to A^{2+} and B^- . The boiling point of a 10.0 molal aqueous solution of AB_2 is _____ $^{\circ}\text{C}$. (Round off to the Nearest Integer) [Given: Molal elevation constant of water, $K_b = 0.5 \text{ K kg mol}^{-1}$, boiling point of pure water = 100°C] [16 March, 2021 (Shift-I)]
114. 1.22 g of an organic acid is separately dissolved in 100 g of benzene ($K_b = 2.6 \text{ K kg mol}^{-1}$) and 100 g of acetone ($K_b = 1.7 \text{ K kg mol}^{-1}$). The acid is known to dimerize in benzene but remain as a monomer in acetone. The boiling point of the solution in acetone increases by 0.17°C . The increase in boiling point of solution in benzene in $^{\circ}\text{C}$ is $x \times 10^{-2}$. The value of x is _____. (Nearest Integer) [Atomic mass: C = 12.0, H = 1.0, O = 16.0] [31 Aug, 2021 (Shift-II)]
115. 224 mL of $\text{SO}_2(\text{g})$ at 298 K and 1 atm is passed through 100 mL of 0.1 M NaOH solution. The non-volatile solute produced is dissolved in 36 g of water. The lowering of vapour pressure of solution (assuming the solution is dilute) ($P_{(\text{H}_2\text{O})}^{\circ} = 24 \text{ mm of Hg}$) is $x \times 10^{-2}$ mm of Hg, the value of x is _____. (Integer answer) [26 Feb, 2021 (Shift-I)]
116. 2 molal solution of a weak acid HA has a freezing point of 3.885°C . The degree of dissociation of this acid is _____ $\times 10^{-3}$. (Round off to the Nearest Integer). [Given: Molal depression constant of water = $1.85 \text{ K kg mol}^{-1}$ Freezing point of pure water = 0°C] [18 March, 2021 (Shift-I)]
117. When 3.00 g of a substance 'X' is dissolved in 100 g of CCl_4 , it raises the boiling point by 0.60 K. The molar mass of the substance 'X' is _____ g mol^{-1} . (Nearest Integer) [Given: K_b for CCl_4 is $5.0 \text{ K kg mol}^{-1}$] [25 July, 2021 (Shift-II)]
118. C_6H_6 freezes at 5.5°C . The temperature at which a solution of 10 g of C_4H_{10} in 200 g of C_6H_6 freeze is _____ $^{\circ}\text{C}$. (The molal freezing point depression constant of C_6H_6 is 5.12°C/m) [24 Feb, 2021 (Shift-II)]
119. 40 g of glucose (Molar mass = 180) is mixed with 200 mL of water. The freezing point of solution is _____ K. (Nearest Integer) [Given: $K_f = 1.86 \text{ K kg mol}^{-1}$; Density of water = 1.00 g cm^{-3} ; Freezing point of water = 273.15 K] [27 Aug, 2021 (Shift-II)]
- Question stem for Question No. (120-121)**
The boiling point of water in a 0.1 molal silver nitrate solution (solution A) is $x^{\circ}\text{C}$. To this solution A, an equal volume of 0.1 molal aqueous barium chloride solution is added to make a new solution B. The difference in the boiling points of water in the two solutions A and B is $y \times 10^{-2}^{\circ}\text{C}$. (Assume: Densities of the solutions A and B are the same as that of water and the soluble salts dissociate completely. Use: Molal elevation constant (Ebulloscopic constant). $K_b = 0.5 \text{ K kg mol}^{-1}$; Boiling point of pure water as 100°C .)
120. The value of x is _____. [JEE Adv 2021]
121. The value of |y| is _____. [JEE Adv 2021]
122. On dissolving 0.5 g of a non-volatile non-ionic solute to 39 g of benzene, vapor pressure decreases from 650 mm Hg to 640 mm Hg. The depression of freezing point of benzene (in K) upon addition of the solute is _____. (Given data: Molar mass and the molal freezing point depression constant of benzene are 78 g mol^{-1} and $5.12 \text{ K kg mol}^{-1}$, respectively) [JEE Adv (2019)]

123. A set of solutions is prepared using 180 g of water as a solvent and 10 g of different non-volatile solutes A, B and C. The relative lowering of vapour pressure in the presence of these solutes are in the order [Given, molar mass of A = 100 g mol⁻¹; B = 200 g mol⁻¹; C = 10,000 g mol⁻¹]
[6 Sept, 2020 (Shift-II)]
- (a) A > C > B (b) C > B > A
(c) A > B > C (d) B > C > A
124. The size of a raw mango shrinks to a much smaller size when kept in a concentrated salt solution. Which one of the following processes can explain this?
[2 Sept, 2020 (Shift-II)]
- (a) Osmosis (b) Reverse osmosis
(c) Diffusion (d) Dialysis
125. How much amount of NaCl should be added to 600 g of water ($\rho = 1.00 \text{ g/mL}$) to decrease the freezing point of water to -0.2°C _____?
(The freezing point depression constant for water = 2 K kg mol⁻¹)
[9 Jan, 2020 (Shift-I)]
126. If 250 cm³ of an aqueous solution containing 0.73 g of a protein A is isotonic with one litre of another aqueous solution containing 1.65 g of a protein B, at 298 K, the ratio of the molecular masses of A and B is _____ $\times 10^{-2}$ (to the nearest Integer).
[3 Sept, 2020 (Shift-I)]
127. The elevation of boiling point of 0.10 m aqueous CrCl₃·xNH₃ solution is two times that of 0.05 m aqueous CaCl₂ solution. The value of x is _____.
[Assume 100% ionisation of the complex and CaCl₂, coordination number of Cr as 6, and that all NH₃ molecules are present inside the coordination sphere]
[6 Sept, 2020 (Shift-I)]
128. The osmotic pressure of a solution of NaCl is 0.10 atm and that of a glucose solution is 0.20 atm. The osmotic pressure of a solution formed by mixing 1 L of the sodium chloride solution with 2 L of the glucose solution is $x \times 10^{-3}$ atm. x is _____. (nearest Integer)
[4 Sept, 2020 (Shift-II)]
129. Molecules of benzoic acid (C₆H₅COOH) dimerise in benzene. 'W'g of the acid dissolved in 30 g of benzene shows a depression in freezing point equal to 2 K. If the percentage association of the acid to form dimer in the solution is 80, then W is
(Given that K_f = 5 K kg mol⁻¹, Molar mass of benzoic acid = 122 g mol⁻¹)
[12 Jan, 2019 (Shift-II)]
- (a) 1.5 g (b) 2.4 g
(c) 1.8 g (d) 1.0 g
130. The freezing point of diluted milk sample is found to be 0.2°C, while it should have been -0.5°C for pure milk. How much water has been added to pure milk to make the diluted sample?
[11 Jan, 2019 (Shift-I)]
- (a) 1 cup of water to 2 cups of pure milk
(b) 3 cups of water to 2 cups of pure milk
(c) 1 cup of water to 3 cups of pure milk
(d) 2 cups of water to 3 cups of pure milk
131. Freezing point of a 4% aqueous solution of X is equal to freezing point of 12% aqueous solution of Y. If molecular weight of X is A, then molecular weight of Y is _____
[12 Jan, 2019 (Shift-I)]
- (a) 2A (b) 3A
(c) A (d) 4A
132. K₂Hgl₄ is 40% ionised in aqueous solution. The value of its van't Hoff factor (i) is _____
[11 Jan, 2019 (Shift-II)]
- (a) 1.6 (b) 2.0
(c) 2.2 (d) 1.8
133. The osmotic pressure of a dilute solution of an ionic compound XY in water is four times that of a solution of 0.01 M BaCl₂ in water. Assuming complete dissociation of the given ionic compounds in water, the concentration of XY (in mol L⁻¹) in solution is _____
[9 April, 2019 (Shift-I)]
- (a) 4×10^{-4} (b) 6×10^{-2}
(c) 16×10^{-4} (d) 4×10^{-2}
134. Molal depression constant for a solvent is 4.0 K Kg mol⁻¹. The depression in the freezing point of the solvent for 0.03 mol kg⁻¹ solution of K₂SO₄ is: (Assume complete dissociation of the electrolyte)
[9 April, 2019 (Shift-II)]
- (a) 0.12 K (b) 0.36 K
(c) 0.18 K (d) 0.24 K
135. 1 g of non-volatile non-electrolyte solute is dissolved in 100g of two different solvents A and B whose ebullioscopic constant are in the ratio of 1 : 5. The ratio of the elevation in their boiling points, $\frac{\Delta T_b(A)}{\Delta T_b(B)}$ is:
[10 April, 2019 (Shift-II)]
- (a) 5 : 1 (b) 10 : 1
(c) 1 : 5 (d) 1 : 0.2
136. A solution containing 62 g ethylene glycol in 250 g water is cooled to -10°C . If K_f for water is 1.86 K kg mol⁻¹, the amount of water(in g) separated as ice is _____
[9 Jan, 2019 (Shift-II)]
- (a) 48 (b) 32
(c) 64 (d) 16
137. A solution is prepared by dissolving 0.6 g of urea (molar mass = 60 g mol⁻¹) and 1.8 g of glucose (molar mass = 180 g mol⁻¹) in 100 mL of water at 27°C. The osmotic pressure of the solution is : _____
[12 April, 2019 (Shift-II)]
(R = 0.08206 L atm K⁻¹ mol⁻¹)
- (a) 8.2 atm (b) 1.64 atm
(c) 4.92 atm (d) 2.46 atm

138. Elevation in the boiling point for 1 molal solution of glucose is 2 K. The depression in the freezing point for 2 molal solution of glucose in the same solvent is 2 K. The relation between K_b and K_f is [10 Jan, 2019 (Shift-II)]
- (a) $K_b = 0.5 K_f$ (b) $K_b = 2 K_f$
(c) $K_b = 1.5 K_f$ (d) $K_b = K_f$
139. On dissolving 0.5 g of a non-volatile non-ionic solute to 39 g of benzene, vapor pressure decreases from 650 mm

Hg to 640 mm Hg. The depression of freezing point of benzene (in K) upon addition of the solute is _____.

(Given data: Molar mass and the molal freezing point depression constant of benzene are 78 g mol^{-1} and $5.12 \text{ K kg mol}^{-1}$, respectively) [JEE Adv 2019]

Exercise -3 (PYQ's)

- | | | | | | | | | | |
|---------------|---------------------|---------------------|------------|------------|---------------------|------------|----------------|-------------|--------------|
| 1. (d) | 2. (c) | 3. (d) | 4. [364] | 5. [180] | 6. [5] | 7. [139] | 8. [36] | 9. [15] | 10. [221] |
| 11. (c) | 12. (a) | 13. [54] | 14. [20] | 15. [64] | 16. [9] | 17. [85] | 18. [3] | 19. [2] | 20. [64] |
| 21. [5] | 22. [13] | 23. [4] | 24. [50] | 25. (c) | 26. [14] | 27. [10] | 28. [100] | 29. [10] | 30. [47] |
| 31. [25] | 32. (c) | 33. (c) | 34. (c) | 35. (b) | 36. (c) | 37. (c) | 38. [2.8-3.05] | 39. [13] | 40. [1111] |
| 41. [82] | 42. [48] | 43. [314] | 44. [555] | 45. [1222] | 46. [2] | 47. [14] | 48. [1] | 49. [25] | 50. [25] |
| 51. [78] | 52. [1] | 53. (d) | 54. (a) | 55. (c) | 56. (d) | 57. [600] | 58. [0.20] | 59. (b) | 60. (d) |
| 61. (a) | 62. (d) | 63. (a) | 64. (b) | 65. (d) | 66. [0.20] | 67. (b) | 68. (d) | 69. (d) | 70. (a) |
| 71. (a) | 72. [240] | 73. [41500] | 74. [4.00] | 75. [100] | 76. [4] | 77. [243] | 78. [372] | 79. [62250] | 80. [8] |
| 81. [40535] | 82. [23] | 83. [30] | 84. [75] | 85. [116] | 86. [682] | 87. (b) | 88. (d) | 89. (c) | 90. (b) |
| 91. [98] | 92. [54] | 93. [8] | 94. [5] | 95. [3] | 96. [80] | 97. [1] | 98. [6] | 99. [15] | 100. [415] |
| 101. (b) | 102. (b) | 103. [100] | 104. [125] | 105. [4] | 106. [269] | 107. [2] | 108. [518] | 109. [36] | 110. [3] |
| 111. [15] | 112. [375] | 113. [106] | 114. [13] | 115. [18] | 116. [50] | 117. [250] | 118. [1] | 119. [271] | 120. [100.1] |
| 121. [2.5] | 122. [0.97 to 1.06] | | 123. (c) | 124. (a) | 125. [1.74 to 1.76] | | 126. [177] | 127. [5.00] | |
| 128. [167.00] | | 129. (b) | 130. (b) | 131. (b) | 132. (d) | 133. (b) | 134. (b) | 135. (c) | 136. (c) |
| 137. (c) | 138. (b) | 139. [0.97 to 1.06] | | | | | | | |